

TEMPORAL SKILLS ARE A CAUSAL BOTTLENECK IN EVOLVING TIME-SERIES AGENTS

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ABSTRACT

Agents operating over evolving time-series data repeatedly face the same analytical structure under new release periods. TimeSage-EV shows that current agents still exhibit recurring temporal-validity failures and recurring exogenous-context failures in this setting. The benchmark’s self-evolving TimeSage-1.0 harness adds a reusable skill library and improves aggregate performance while reducing token cost to $0.82\times$ the baseline. We study the mechanism behind these gains. Our hypothesis is that a large class of recurring failures is caused by missing reusable procedures for cutoff-aware temporal reasoning and exogenous-context integration. We formulate targeted temporal skills as a causal intervention. The intervention injects a compact procedure for a known failure mode while keeping the future endpoint and evidence fixed. A matched generic skill controls for the benefit of receiving any reusable procedure. A no-skill condition provides the baseline. This design distinguishes targeted mechanism repair from generic retrieval assistance. The source benchmark exposes an unusually clean intervention surface because accepted skills become callable tools starting from the next period. We specify a skill contract for temporal cutoff enforcement. A second contract handles release-to-period alignment. A third contract grounds exogenous events. The resulting protocol predicts that targeted skills will remove matched recurring failures more strongly than generic skills while preserving chronological evaluation. This work advances self-evolving time-series agents from aggregate skill ablation toward mechanism-specific causal evaluation. It also provides a general recipe for testing whether recurring agent failures arise from missing reusable procedures.

1 INTRODUCTION

Evolving time-series analysis requires an agent to revisit the same scenario as new releases arrive. The evidence cutoff changes every period. Historical documents remain useful, yet later information must stay hidden. TimeSage-EV formalizes this setting across 60 institutional scenarios and 1,485 scenario-period questions (Yao et al., 2026).

The benchmark reveals recurring failures that have a procedural character. Agents misuse temporal cutoffs. They struggle to align releases with the correct period. They also miss relevant exogenous context when numerical changes need documentary explanation. These errors recur across release periods even though the required procedure is reusable.

TimeSage-1.0 provides evidence that reusable procedures matter. It augments the aligned baseline harness with a self-evolving skill library. A saved skill becomes callable from the next period. The source paper reports higher overall performance with reusable skills. The largest gains appear on medium and hard scenarios. Skill-enabled evaluation also reduces token cost to $0.82\times$ the reference condition (Yao et al., 2026).

Aggregate skill gains leave the mechanism unresolved. A generic skill library can help through many routes. It can reduce repeated coding effort. It can also stabilize formatting. Our target is narrower and more causal: recurring temporal failures arise because the agent lacks a reusable procedure that encodes the required temporal operation.

This hypothesis yields a direct intervention. Identify an endpoint that exhibits a temporal or exogenous-context failure. Construct a targeted skill that implements the missing procedure. Re-evaluate the

Targeted temporal skills isolate a mechanism behind recurring grounding failures

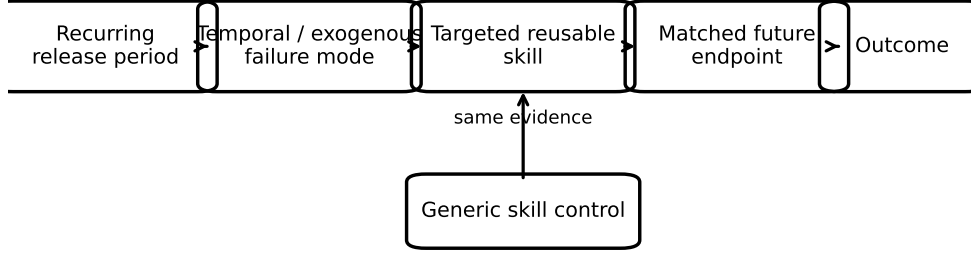


Figure 1: Targeted skill intervention. The same future endpoint is evaluated with a mechanism-specific skill and with a matched generic skill. This isolates the contribution of the procedure encoded by the targeted skill.

same endpoint with the same evidence. Compare against a generic reusable skill that carries similar implementation complexity without encoding the target mechanism. A no-skill condition anchors the baseline.

The TimeSage harness makes this intervention unusually clean. Its Appendix defines an `induct_skill` interface for saving a reusable Python function. It also defines `retrieve_skill` for inspecting saved code. Accepted skills become live tools only in the next period. This chronological rule lets us inject a procedure before a future endpoint without changing the current-period answer.

We develop this design for three recurring mechanism families. One family enforces the information cutoff. Another aligns release dates to analytical periods. The third integrates exogenous events into evidence-grounded reasoning. The paper asks whether targeted procedures causally remove their matched failure modes.

The contribution is a mechanism-specific evaluation of self-evolving skills. TimeSage-EV establishes that reusable skills help in aggregate. We turn the skill library into a causal intervention surface and define matched controls that distinguish targeted mechanism repair from generic assistance. This approach supports a stronger claim about self-evolution: a recurring failure can reveal a missing reusable procedure that should become explicit agent state.

2 RECURRING FAILURE AS A MISSING PROCEDURE

Let a scenario generate a sequence of release periods $t_1, \dots, t_K$. At each period, the agent receives a cutoff-valid evidence state E_t and a question x_t . The agent also carries a reusable skill library L_t . Future behavior follows

$$Y_t = H(x_t, E_t, L_t, S_t), \quad (1)$$

where S_t contains the remaining agent state.

A recurring failure mode appears when the same procedural error repeats across periods while the concrete evidence changes. This pattern suggests that the agent lacks a reusable operation. We represent the missing operation as a skill g_m associated with mechanism m .

The causal estimand compares outcomes on the same future endpoint under two skill libraries:

$$\Delta_m = \mathbb{E}[Y_t \mid do(L_t = L \cup \{g_m\})] - \mathbb{E}[Y_t \mid do(L_t = L \cup \{g_c\})], \quad (2)$$

where g_c is a generic control skill with matched implementation complexity. A positive Δ_m supports the claim that the target procedure repairs the matched failure beyond generic skill assistance.

2.1 TEMPORAL CUTOFF SKILL

The cutoff skill receives a query date and a set of candidate evidence timestamps. It returns the admissible evidence subset under the benchmark cutoff. Its purpose is to prevent later releases from leaking into earlier-period analysis. The skill can also expose the cutoff date explicitly to subsequent reasoning.

2.2 RELEASE ALIGNMENT SKILL

The alignment skill maps source-release dates to the benchmark period represented by the evidence. It is useful when the source publication date differs from the underlying reporting period. The skill returns a normalized period identifier and verifies that the selected time-series row matches that period.

2.3 EXOGENOUS GROUNDING SKILL

The grounding skill links a numerical change to documentary evidence available before the cutoff. It searches prior-period reports for candidate events and returns evidence spans tied to the target period. This procedure supports attribution questions that require context outside the numerical series.

Each skill encodes a reusable operation that excludes scenario-specific answers. The same function can apply across future periods. This property matches the source harness contract for skills saved through `induct_skill`.

3 SOURCE EVIDENCE FOR THE SKILL BOTTLENECK

TimeSage-EV evaluates recurring analysis over institutional releases from February 2023 through May 2026 (Yao et al., 2026). The benchmark reports persistent gaps in temporal validity. It also reports recurring difficulty with exogenous context. These failures remain visible across model tiers, which makes them suitable targets for reusable procedure induction.

The source paper introduces TimeSage-1.0 as a diagnostic self-evolving harness. The base interface remains aligned with the baseline. The skill-library manager is the additional mechanism. Skills are induced from previous reasoning traces and become available to later periods. This gives the benchmark an explicit state-update boundary for testing reusable procedures.

The released aggregate results support the value of this state. Reusable skills improve overall performance, with the clearest gains on medium and hard scenarios. Skill invocation is most frequent on hard scenarios. The source also reports that the skill-enabled condition reduces token cost to $0.82\times$ the reference condition.

These results establish three ingredients for the causal hypothesis. Reusable procedures can enter persistent state. They can influence later periods. They improve aggregate performance while reducing repeated reasoning cost. The remaining question concerns mechanism specificity: which recurring failure families are repaired by which skill?

The benchmark Appendix provides the needed intervention surface. `induct_skill` stores a reusable function with a description and notes. `retrieve_skill` exposes the stored implementation when the short description is insufficient. Skills accepted after period t become callable from period $t + 1$. This chronological registration preserves the evolving evaluation structure.

The source also gives examples of desirable skill granularity. It encourages compact procedures such as schema-specific loaders and date-to-period converters. Scenario-specific answer dictionaries are discouraged. This contract aligns directly with our targeted temporal-skill design.

4 TARGETED SKILL INTERVENTION

The decisive experiment starts from endpoints with a documented recurring failure family. Each endpoint is replayed from the same frozen evidence package under three skill conditions.

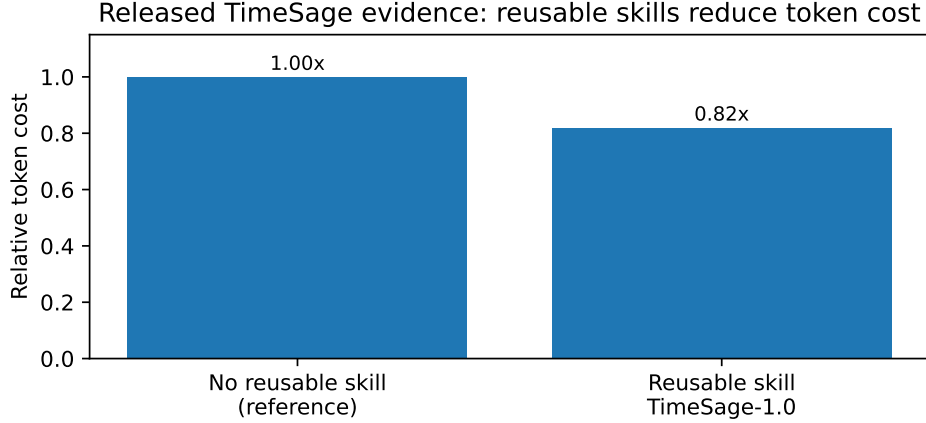


Figure 2: Released TimeSage evidence reports $0.82\times$ relative token cost when reusable skills are enabled. The source also reports higher benchmark performance under the skill-library condition.

The targeted condition injects a skill whose procedure matches the observed failure mechanism. The generic condition injects a frozen target-blind helper under the same callable wrapper and persistent registration timing. The baseline condition adds no experimental skill. Every arm uses the same model configuration; prompt; evidence package; tool interfaces; harness; evaluated period.

4.1 ENDPOINT MATCHING

A temporal-cutoff endpoint is paired with a skill that filters evidence by cutoff time. A release-alignment endpoint receives a date-to-period procedure. An exogenous-grounding endpoint receives a procedure that retrieves prior documentary evidence tied to the numerical change. Endpoint selection is frozen from source failure annotations before outcomes are inspected.

4.2 FROZEN GENERIC-SKILL CONTROL

The generic helper is matched before outcomes are inspected. Its source-token count must be within 10% of the targeted implementation. Its abstract-syntax-tree node count must be within 5%. The maximum number of external tool calls is identical. A static audit rejects the control when it can inspect a target cutoff, map a release date to a reporting period, or select a documentary event as the explanation for a numerical change.

The family-matched generic helper remains useful while target blind. For cutoff endpoints it validates evidence-package schema and data types without reading timestamps. For release-alignment endpoints it checks table and index consistency without reading release dates or reporting-period fields. For exogenous-grounding endpoints it exposes document inventory and format metadata without selecting events or aligning documents to numerical changes. This control absorbs generic benefit from receiving an additional persistent callable tool while excluding the operation whose causal effect is being estimated.

4.3 ORDERED MECHANISM ABLATION

The mechanism test is preregistered as an ordered skill construction. T_1 adds only cutoff-valid evidence filtering. T_2 keeps T_1 unchanged and adds release-to-period normalization with row verification. T_3 keeps T_2 unchanged and adds cutoff-valid documentary grounding for scored attribution targets. Each component source hash is frozen before evaluation. Every targeted arm is paired with the frozen generic control and the no-skill anchor on the same endpoint.

4.4 PRIMARY OUTCOME

The primary metric is correction of the matched failure. A temporal-validity error is corrected when the final answer uses only admissible evidence. A release-alignment error is corrected when the answer references the correct period. An exogenous-context error is corrected when the attribution is supported by cutoff-valid documentary evidence.

The targeted-skill effect is

$$\Delta_{\text{target}} = S_{\text{target}} - S_{\text{generic}}, \quad (3)$$

where S is success on the matched failure criterion. The paired no-skill difference is reported as a baseline anchor. A positive targeted-minus-generic effect identifies the mechanism-specific contribution after generic callable assistance has been matched.

Every result row is immutable and content-addressed. It binds the repository revisions; scenario and period identity; cutoff; failure family; condition; skill source hash; model configuration; harness hash; repeat identifier; raw response hash; family-specific outcome; source-native evaluator payload. This prevents a repair decision from drifting away from the exact benchmark object that produced it.

4.5 TRANSFER TEST

A reusable procedure should generalize beyond the endpoint that motivated it. The first transfer test applies the frozen skill without edits to later periods from the same scenario under the same generic and no-skill controls. A compatible cross-domain test follows only when the first-party interface exposes the same registered failure-family criterion. The transferred skill cannot contain scenario-specific facts or receive outcome-dependent edits.

4.6 CHRONOLOGICAL DEPLOYMENT

The intervention respects the TimeSage chronological contract. A skill enters the library before the evaluated future period. It cannot modify the acquisition-period answer. This preserves the direction of self-evolution and prevents current-answer leakage.

The experiment turns a recurring error into a falsifiable memory hypothesis. If a targeted skill fails to outperform the generic control, the missing-procedure explanation loses support for that failure family. If the effect transfers to later periods, the skill becomes direct evidence that the recurring failure was caused by a missing reusable operation.

5 RELATED WORK

Evolving time-series agents. TimeSage-EV introduces recurring period-level evaluation with strict temporal cutoffs and traceable source evidence (Yao et al., 2026). Its TimeSage-1.0 harness adds a self-evolving skill library and demonstrates aggregate gains from reusable procedures. Our work moves from the aggregate ablation to mechanism-specific causal intervention.

Experience-derived agent memory. Agent Workflow Memory induces reusable procedures from prior trajectories (Wang et al., 2024). ReasoningBank stores reusable reasoning memories derived from experience (Ouyang et al., 2026). These systems show that structured experience can improve later tasks. The targeted-skill protocol studies which specific missing procedure explains a recurring failure.

Skill libraries as intervention surfaces. A skill library exposes a modular state boundary. One procedure can be added while the rest of the agent remains fixed. This supports causal tests that are harder to perform with a single undifferentiated conversation memory. TimeSage’s chronological skill registration further protects the direction of the intervention because a new skill affects only later periods.

6 LIMITATIONS

The TimeSage paper declares a public dataset and code release. A fresh 2026-08-22 first-party audit finds GitHub master at 41f5e7656cb8 with only README.md; the README ends with Upcoming.... The Hugging Face dataset is public at revision 711f6443dc76 with only .gitattributes. The frozen May 2026 evaluated snapshot is therefore unavailable from the declared release points. Targeted reruns remain support debt until the evaluated snapshot or equivalent first-party artifacts are available.

The source paper reports aggregate gains from reusable skills and a $0.82\times$ token-cost ratio. It does not report the targeted intervention proposed here. The mechanism-specific causal effect therefore remains experiment debt even though the aggregate skill benefit is already observed.

Failure-family labels must be frozen before intervention outcomes are inspected. Temporal-validity cases require evidence-cutoff errors that can be checked against the benchmark timestamp contract. Exogenous-context cases require a source-grounded attribution target. This annotation pass remains experiment debt for the frozen snapshot.

A targeted skill can change more than one internal reasoning step. The matched generic-skill control reduces this concern through a preregistered target-blind helper. Callable signature and persistent timing are exact matches. Source-token count is matched within 10%, abstract-syntax-tree node count within 5%, and maximum external tool-call count exactly. The remaining uncertainty is behavioral: the released harness must show that this generic helper supplies comparable reusable assistance without performing the target operation.

The paper focuses on reusable procedural state. Some recurring failures can arise from model capability limits even when the correct procedure is available. The targeted-skill effect therefore estimates the subset of failures that are repairable through explicit reusable procedures.

7 CONCLUSION

TimeSage-EV exposes recurring temporal failures in an evolving analytical setting. Its self-evolving harness shows that reusable skills improve aggregate performance and reduce token cost. These observations support a stronger causal hypothesis: a substantial class of recurring failures is produced by missing reusable temporal procedures.

We formulate a targeted skill intervention that keeps the future endpoint fixed and compares a mechanism-specific skill against a matched generic helper. The design applies cleanly to cutoff enforcement, release alignment and exogenous grounding through separate skill contracts. This turns failure analysis into a direct self-evolution mechanism. A recurring failure identifies a candidate procedure, and the future period tests whether adding that procedure repairs the agent state.

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A TARGETED SKILL CONTRACTS

A.1 TEMPORAL CUTOFF CONTRACT

Input: target period and candidate evidence timestamps. Output: the subset that is admissible before the benchmark cutoff. The function also returns the cutoff timestamp used for filtering. The skill contains no scenario answer.

A.2 RELEASE ALIGNMENT CONTRACT

Input: publication date and source reporting metadata. Output: the benchmark period represented by the release. The procedure validates the selected time-series row against the normalized period key.

A.3 EXOGENOUS GROUNDING CONTRACT

Input: target period and numerical change description. Output: cutoff-valid documentary evidence spans that can support attribution. The procedure returns evidence identifiers and short excerpts for later reasoning.

A.4 MATCHED GENERIC CONTROLS

Each targeted skill is paired with a frozen target-blind helper. The callable signature and persistent registration timing are identical. Source-token count is matched within 10%. Abstract-syntax-tree node count is matched within 5%. The maximum external tool-call count is identical. The cutoff-family control validates package schema without reading timestamps. The release-family control checks table and index consistency without reading release dates. The grounding-family control exposes document inventory metadata without selecting events or aligning documents to numerical changes.

A.5 RESULT-ROW BINDING

Each endpoint-condition observation binds the dataset revision and code revision to the scenario-period identity. It also binds the cutoff, failure family and skill source hash. Model configuration and harness hashes remain attached to the raw response hash. Family success and the source-native evaluator payload are recorded in the same immutable row.

A.6 FROZEN SUPPORT STATE

At the 2026-08-22 repair audit, the first-party TimeSage-EV GitHub master head was 41f5e7656cb8; its root tree contained only README.md, whose public content ended with Upcoming. . . . The public Hugging Face dataset revision was 711f6443dc76; its root tree contained only .gitattributes. These revisions preserve the exact external-support boundary used by the intervention gate.